

Explicit formula for the $(\mathrm{GL}_2, \mathrm{GL}_2)$ theta lift via Bruhat decomposition

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ABSTRACT. We explicitize the algebraic approach to the $(\mathrm{GL}_2(\mathbb{Q}), \mathrm{GL}_2(\mathbb{Q}))$ -theta correspondence (of [BCG3]) using theta series in the distributional de Rham complex of the squared universal elliptic curve. As a result, we obtain novel *unsmoothed* formulas for this theta lift, analogous to Siegel’s classical formulas for periods of Eisenstein series. In the setting of CM elliptic curves, we obtain a $(\mathrm{GL}_2(K), K^\times)$ theta lift formula, which earlier was obtained analytically by Ito. We discuss some applications to formulas for L -values.

1. INTRODUCTION

In the article [BCG1] and its sequel [BCG3], the authors construct a theta lift from the $(n - 1)$ -homology of GL_n to modular forms for GL_2 . This construction uses an equivariant polylogarithm class; in particular, when $n = 2$, the class in question corresponds to the algebraic GL_2 -action on the squared universal elliptic curve over the upper half-plane (or its modular curve quotients). This class can be computed by various means: in [BCG1], the lifts of certain modular geodesics are computed via the seesaw principle, while in [BCG3], certain stabilizations of the lift are computed by relating them to values “at the boundary,” i.e. by partial modular symbols. Related cocycles valued in K -theory were constructed in [SV] and [BPPS] (for general n ; the regulators of these cocycles correspond to parabolic stabilizations of the theta lift.

In [X4], we constructed explicit elements in the GL_2 -equivariant distributional de Rham complex of a torus to re-derive classical “unstabilized” formulas for periods of weight-2 Eisenstein series. In cohomology, these Eisenstein series arise as pullbacks of an equivariant class on $(S^1)^2$, allowing us to compute them by explicating equivariant cohomology via geometrically-defined complexes. In this note, we apply the same technique to the squared universal elliptic curve, and in turn obtain novel explicit formulas for the $(\mathrm{GL}_2, \mathrm{GL}_2)$ -theta lift. These formulas especially shed light on the Eisenstein component of the lift, exhibiting the particular combination of weight-2 Eisenstein series which arise from specializations at parabolic matrices.

Due to the nature of the theta lift’s construction, instead of the equivariant distributional de Rham complex of [X4], we will use the equivariant weight-2 distributional *Dolbeault* complex of our squared universal elliptic curve. The explicit elements in loc. cit. were comprised of weight-1 and weight-2 Bernoulli polynomials $B_1(z)$ and $B_2(z)$; these will be replaced with theta series $E_1(\tau, z)$ and $E_2(\tau, z)$ interpolating weight-1 and 2 Eisenstein series. To find the explicit forms of these elements, as in loc. cit., we will need to use an explicit version of the Bruhat decomposition of

$$\mathrm{GL}_2(\mathbb{Q}) = BwB$$

where

$$w = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

represents a Weyl element, and $B \subset \mathrm{GL}_2(\mathbb{Q})$ is the upper triangular Borel subgroup. In particular, we recover the following explicit formula:

Theorem 1.1 (Theorem 3.8). *Let $\Gamma \subset \mathrm{GL}_2(\mathbb{Z})$ be an arithmetic subgroup, and consider any Γ -fixed linear combination*

$$D = \sum_i c_i [(u_i, v_i)],$$

of nonzero torsion sections (u_i, v_i) of the squared universal elliptic curve $E \times_{\mathcal{H}} E$ over the upper half-plane, endowed with the block scalar action of $M_2(\mathbb{Z})$. Let

$$\Theta_D^{(c)} : \Gamma \rightarrow \text{modular forms of weight 2 and level } H$$

be the associated Eisenstein theta lift for $(\mathrm{GL}_2, \mathrm{GL}_2)$ defined by (2.5) (summed over $x = (u_i, v_i)$ with multiplicities c_i), where here $H \leq \mathrm{SL}_2(\mathbb{Z})$ is some level structure fixing the torsion cycle D . Define a function parameterized by elements $\gamma \in \mathrm{GL}_2(\mathbb{Q})$ by the formula

(1.1)

$$\theta[\gamma](\tau, z_1, z_2) := \begin{cases} \frac{4b}{d} E_2(\tau, z_2) & \text{if } c = 0, \text{ else} \\ \begin{pmatrix} a & 1 \\ c & 0 \end{pmatrix}_* \frac{4d}{c \det \gamma} E_2(\tau, z_2) + \frac{1}{c} \begin{pmatrix} 1 & a \\ 0 & c \end{pmatrix}_* E_1(\tau, z_1) E_1(\tau, z_2) + \frac{4a}{c} E_2(\tau, z_2) \end{cases}$$

where E_1 and E_2 are specializations in weights 1 and 2 of the classical Kronecker-Eisenstein series, defined in Section 3.1. Then we have

$$\Theta_D^{(c)}(\gamma) = \sum_i c_i \theta[\gamma](\tau, u_i, v_i) \in \mathbb{C}.$$

Here, c_i are integer coefficients, and u_i and v_i are elements of $(\mathbb{Z}/N)^2 - \{(0, 0)\}$, thought of as N -torsion sections.

In Section 4.3, we outline also that the same methods yield an almost identical formula (4.1) for subgroups of $\mathrm{GL}_2(K)$ for K an imaginary quadratic field, valued in CM specializations (for the field K) of E_1 and E_2 . This formula was previously obtained analytically by Ito [Ito], who related it also to L -values of the corresponding CM fields.

Correspondingly, by evaluating the $(\mathrm{GL}_2(\mathbb{Q}), \mathrm{GL}_2(\mathbb{Q}))$ theta lift on primitive hyperbolic matrices, a seesaw diagram (as in [BCG1, §13]) shows that we obtain formulas for diagonal restrictions of Hilbert-Eisenstein series:

Theorem 1.2 (Theorem 4.3). *In the notation of [DDP, §2.2], let $G_1(1, 1_{[a]}^{-(p)})$ be the diagonal restriction of a weight-1 Hilbert-Eisenstein series for a real quadratic field F , corresponding to the p -stabilized indicator*

function of a wide ideal class $[\mathfrak{a}]$. Then we have the identity of level- $\Gamma_0(p)$ modular forms

$$G_1(1, 1_{[\mathfrak{a}]}^{-(p)}) = \frac{d_F}{p^2[\mathcal{O}_F^\times : U_p]} \sum_{x \in E[p]^2 - K_p^2} \left[\frac{4(a+d)}{c} E_2(\tau, x_2) + \frac{1}{c} \left\{ \begin{pmatrix} 1 & a \\ 0 & c \end{pmatrix}_* E_1(\tau, z_1) E_1(\tau, z_2) \right\}_{z=x} \right]$$

where E_1 and E_2 are classical weight-1 and 2 Eisenstein series as before, K_p is a $\Gamma_0(p)$ -fixed combination of torsion sections of the universal elliptic curve, a, b, c, d are the entries of a primitive hyperbolic matrix in $\mathrm{SL}_2(\mathbb{Z})$ stabilizing an indefinite quadratic form corresponding to $[\mathfrak{a}]$, U_p is the group of norm-1 units of F which are 1 modulo p , and d_F is the discriminant of F .

1.1. Related work. A similar computation of equivariant cohomology was completed in [KS, §3], using generalized Kronecker-Eisenstein series. Our calculation can be considered as a more explicated form of the calculation there, for the self-product of the universal elliptic curve with trivial coefficients, except that:

- (1) we treat the entire action of GL_2 at once, instead of just working with the equivariance under one subtorus at a time;
- (2) we construct also a cocycle for a larger group with no geometric action on the fibers ($\mathrm{GL}_2(\mathbb{Q})$ instead of $\mathrm{GL}_2(\mathbb{Z})$, or $\mathrm{GL}_2(K)$ in the CM case);
- (3) we do not rely on any stabilization or smoothing coming from translations by torsion sections.

We restrict to this setting both because it is particularly arithmetically rich, and allows for a considerably simpler and less technical exposition. Furthermore, by eschewing smoothing, the resulting formulas are more transparently related to classical objects such as classical holomorphic Eisenstein series or the period formulas of Siegel (as in [X4]), and corresponds directly to the theta kernel of integration of [BCG1]. Furthermore, our formulas can be specialized (with cohomological meaning) at arbitrary non-zero torsion points, thanks to Theorem 3.6. As we noted in our previous article [X4] in a simpler setting, our same methods should produce analogous formulas with twisted coefficients corresponding to higher-weight automorphic sheaves; the resulting formulas will involve progressively more complicated combinations of higher-weight specializations of Kronecker-Eisenstein series. Another natural generalization would be to use the more complicated decomposition of parabolic subgroups for GL_n and n -fold products of elliptic curves; in this setting, our unsmoothed formulas grow in complexity very quickly with n .

The $(\mathrm{GL}_2, \mathrm{GL}_2)$ -theta lift is closely related to Gross-Stark units for real quadratic fields, and to Stark-Heegner points, via the p -adic Kudla program approach of [DPV1]; this is the subject of work in preparation, both independently and with Marti Roset Julia. However, in these works, we always use smoothed versions of the formulas, since they are easier to describe in more generality, and result in p -adic interpolability.

2. EISENSTEIN THETA LIFT FOR $(\mathrm{GL}_2, \mathrm{GL}_2)$

The theta lift we are considering was originally constructed purely analytically in [BCG1, §13] (the case $N = 2$ there), but we will find the algebraic construction of [BCG3, Thm. 1.8] more useful; the comparison between the two is also proven in Chapitre 9 of loc. cit.

To begin, we consider the complex upper half-plane $\mathcal{H}$, and the universal elliptic curve over it given by the complex uniformization

$$E := (\mathcal{H} \times \mathbb{C}) / \{(\tau, \mathbb{Z} + \tau\mathbb{Z})\}.$$

Here, we will write $\tau = x + yi$ for the coordinate on the base, and z for the fiberwise coordinate on $\mathbb{C}$. This fiber bundle has the complex analytic action (as a fiber bundle) of $\mathrm{GL}_2(\mathbb{Z})$ by

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} (\tau, z) = \left(\frac{a\tau + b}{c\tau + d}, \frac{z}{c\tau + d} \right).$$

We will also write $z = u - \tau v$ for $u, v \in \mathbb{R}$; this then allows $\mathrm{GL}_2(\mathbb{Z})$ to act on the column vector (u, v) by the standard left action. The principal space we will work on is the self-product $T := E \times_{\mathcal{H}} E$, which we will coordinatize by τ on the base, and z_1 and z_2 on the two elliptic curves. For either of these bundles, we denote by $\pi : T \rightarrow \mathcal{H}$ or $\pi : E \rightarrow \mathcal{H}$ the projection maps.

For us, the important property of T is that it has a fiberwise action by endomorphisms of $M_2(\mathbb{Z})$, coming from the endomorphism action of $\mathbb{Z}$ on any elliptic scheme by isogenies, i.e. the integer $a \in \mathbb{Z}$ acts by the finite multiplication-by- a isogeny $[a]_*$ of degree a^2 . In particular, the group $\mathrm{GL}_2(\mathbb{Z})$ acts by automorphisms on T .

Remark 2.1. For any torsion-free congruence subgroup $H \subset \mathrm{SL}_2(\mathbb{Z})$, we may quotient the bundle $\pi : E \rightarrow \mathcal{H}$ by H (by the matrix action indicated above) to obtain the universal elliptic curve $E(H)$ over the modular curve $Y(H)$ of level H , the latter of which is a fine moduli space; we may also form $T(H) := E(H) \times_{Y(H)} E(H)$. In this article, for technical reasons, rather than working on this level, we make all our constructions at the level of the bundles T and E over $\mathcal{H}$ rather than descending to the quotient, observing the H -invariance only at the end.

2.1. Dolbeault complexes and equivariant cohomology. We will follow the naive “Borel” approach to equivariant cohomology used in, e.g., [KS, Appendix A] and [BCG3, Annexe A], which is computed by forming the double complex of group cochains from any suitably functorial complex computing the (non-equivariant) cohomology theory. In this section, we recall some facts about our primary cohomology complex of interest: *distributional Dolbeault complex*, computing coherent cohomology. Our reference for the various facts about Dolbeault resolutions below is [Dem, Chapter 5].

Let X be a complex manifold of complex dimension d , and write $\mathcal{A}_X^{p,q}$ for the sheaves of smooth complex differentials on X of holomorphic degree p and antiholomorphic degree q ; these sheaves are flabby on X ,

so we (a bit abusively) use this same notation for their global sections. These forms support the pair of anti-commuting Dolbeault differentials

$$\partial : \mathcal{A}_X^{p,q} \rightarrow \mathcal{A}_X^{p+1,q}, \bar{\partial} : \mathcal{A}_X^{p,q} \rightarrow \mathcal{A}_X^{p,q+1}$$

which are square zero, satisfy the Leibniz rule, and sum to the usual exterior derivative on differential forms. For an integer $w \geq 0$, we recall that sheaf of holomorphic $(w, 0)$ -differentials Ω_X^w on X is precisely the kernel of $\bar{\partial}$ inside $\mathcal{A}_X^{w,0}$. Ordinarily, the Dolbeault resolution of Ω_X^2 is constructed from these sheaves, but instead, we dualize: first, write $\mathcal{A}_{X,c}^{p,q}$ for the space of *compactly supported* relative (p, q) -differentials on X , carrying the same differentials as the usual ones (though they do not form a sheaf). Then we define the continuous dual¹

$$\mathcal{D}_X^{p,q} = \text{hom}_{\text{cont}}(\mathcal{A}_{X,c}^{d-p,d-q}, \mathbb{C})$$

which we call the space of (p, q) -currents on X . Unlike the compactly supported forms, these *do* form a sheaf, since they can be restricted along inclusions of opens in adjunction to the pushforward of compactly supported forms. Each $\mathcal{D}_X^{p,q}$ is soft, and so we abusively use the same notation for the sheaf and its global sections.

The Dolbeault operators

$$\partial : \mathcal{D}_X^{p,q} \rightarrow \mathcal{D}_X^{p+1,q}, \bar{\partial} : \mathcal{D}_X^{p,q} \rightarrow \mathcal{D}_X^{p,q+1}$$

are defined as the graded adjoints of the exterior derivative on forms, i.e.

$$(\partial c)(\eta) := (-1)^{\deg c} c(\partial \eta), (\bar{\partial} c)(\eta) := (-1)^{\deg c} c(\bar{\partial} \eta)$$

where $\deg$ refers to the total degree (i.e. $p + q$ for a (p, q) -current).

There is a natural inclusion

$$\nu_X : \mathcal{A}_X^{p,q} \rightarrow \mathcal{D}_X^{p,q}$$

given by

$$\omega \mapsto \left(\eta \mapsto \int_X \omega \wedge \eta \right)$$

where η is a compactly supported smooth $(d - p, d - q)$ -form; the integral makes sense and is finite since the wedge product is also compactly supported, and of degree (d, d) . It is clear that ν_X commutes with the Dolbeault differentials ∂ and $\bar{\partial}$. Furthermore, since (p, q) -compactly supported forms have covariant functoriality by flat holomorphic maps and contravariant by proper holomorphic maps, by adjunction, (p, q) -relative currents can be pulled back by flat maps and pushed forward by proper ones. One can check that the push-pull formula for differential implies that ν_X is functorial for flat pullback and proper pushforward of complex manifolds.²

¹The topology is the inductive limit topology on compactly supported smooth forms coming from C^∞ -topologies of uniform convergence on compact subsets of X .

²For orientation-reversing maps, one has to be careful about this comparison map, since they introduce an extra sign; this was an issue we had to work with in [X4]. However, all of our maps will be morphisms of complex manifolds, and hence orientation-preserving.

There is a $\bar{\partial}$ -Poincaré (or Dolbeault-Grothendieck) lemma [Skoda] for these currents. As with the classical Poincaré lemma, these are cohomologically useful when stated in the following form:

Lemma 2.2. *The complex*

$$(2.1) \quad \Omega_X^w \hookrightarrow \mathcal{D}_X^{w,0} \xrightarrow{\bar{\partial}} \mathcal{D}_X^{w,1} \xrightarrow{\bar{\partial}} \dots$$

is an acyclic resolution of sheaves. Further, the map of $\bar{\partial}$ -complexes induced by ν_X

$$(2.2) \quad \mathcal{A}_X^{w,\bullet} \rightarrow \mathcal{D}_X^{w,\bullet}$$

is a quasi-isomorphism for each degree w .

In particular, for 2-dimensional X and $w = 2$, the complex

$$\mathcal{D}_X^{2,0} \xrightarrow{\bar{\partial}} \mathcal{D}_X^{2,1} \xrightarrow{\bar{\partial}} \mathcal{D}_X^{2,2}$$

is functorial in X (in the sense described previously) and computes the coherent cohomology of Ω_X^2 .

An important class of currents is those associated to closed complex submanifolds $Z \subset X$ of codimension r : these are the *current of integration*

$$\delta_Z \in \mathcal{D}_X^{r,r}$$

defined by

$$\delta_Z(\omega) := \int_Z \omega.$$

This is a special case of the more general phenomenon that for a codimension- r closed immersion of closed manifolds $i : Z \rightarrow X$, pullback of compactly supported forms induces a pushforward

$$i_* : \mathcal{D}_Z^{a,b} \rightarrow \mathcal{D}_X^{a+r,b+r}$$

The pushforward of the constant function (viewed as a $(0,0)$ -current) yields the current of integration of Z .

Letting $j : \mathcal{D}_X^{w,\bullet} \rightarrow \mathcal{D}_{X-Z}^{w,\bullet}$ be the restriction map to the complementary open, we can functorially form the mapping cone $Cj^{w,\bullet} := \text{cone}(\mathcal{D}_X^{w,\bullet} \rightarrow \mathcal{D}_{X-Z}^{w,\bullet})$, which in degree k is $\mathcal{D}_X^{w,k} \oplus \mathcal{D}_{X-Z}^{w,k-1}$, and whose cohomology we denote by $H_Z^\bullet(X, \Omega_X^w)$ (and equivariant cohomology similarly with a group subscript). The tautological exact triangle

$$(2.3) \quad Cj^{w,\bullet} \rightarrow \mathcal{D}_X^{w,\bullet} \rightarrow \mathcal{D}_{X-Z}^{w,\bullet}$$

then gives rise to functorial long exact localization sequences in both usual and equivariant cohomology.

There is an evident map $\mathcal{D}_Z^{w,\bullet} \rightarrow Cj^{w+r,\bullet+r}$, $\eta \mapsto (i_*\eta, 0)$, which one might hope to be a “purity” quasi-isomorphism, but this fails in general. However, the pushforward map

$$\mathcal{D}_Z^{w,\bullet} \xrightarrow{i_*} \mathcal{D}_X^{w+r,\bullet+r}$$

visibly factors as a composite

$$\mathcal{D}_Z^{w,\bullet} \rightarrow Cj^{w+r,\bullet+r} \rightarrow \mathcal{D}_X^{w+r,\bullet+r}.$$

The constructions of this section also have analogues for de Rham cohomology, forgetting about the separation of holomorphic/anti-holomorphic degrees: for instance, we have resolution of the constant sheaf by soft sheaves of *all* currents of nonnegative integer degrees

$$\mathbb{R} \rightarrow (\mathcal{D}_X^0 \xrightarrow{d} \mathcal{D}_X^1 \xrightarrow{d} \dots)$$

on any smooth manifold X , functorial also for finite pushforward and flat pullback; consequently, the complex in parentheses computes the real de Rham cohomology of X . A main difference to note is that the local pushforward yields an excision (or “purity”) isomorphism

$$H^{i-\text{codim } Z}(Z) \xrightarrow{\sim} H_Z^i(X),$$

in this setting, so the analogue to (2.3),

$$(2.4) \quad \mathcal{D}_Z^\bullet \xrightarrow{f_*} \text{cone}(\mathcal{D}_X^{\bullet+\text{codim } Z} \rightarrow \mathcal{D}_{X-Z}^{\bullet+\text{codim } Z}),$$

is a quasi-isomorphism, unlike in the coherent cohomology setting.

2.2. Eisenstein classes in equivariant cohomology. Let $\Gamma \subset \text{GL}_2(\mathbb{Z})$ be any subgroup; we will work with the Γ -equivariant cohomology of Γ -subspaces of T . For example, thanks to the previous section, the coherent cohomology of Ω_T^2 we may compute by the double complex $C^\bullet(\Gamma, \mathcal{D}_T^{2,\bullet})$.

In this article, we use the construction of [KS, §2.3] of an “equivariant Eisenstein class” in coherent cohomology of relative sheaves of differentials (in our case, for the bundle T over the upper half-plane $\mathcal{H}$). We then briefly recall the arguments from the author’s thesis [X, §3] and Bergeron–Charollois–Garcia [BCG3, §9] which relate this algebraic construction to the analytic theta lift of the latter authors, constructed in [BCG1, §13].

For $a \in \mathbb{Z}$, we denote by $[a] : T \rightarrow T$ the associated scalar multiplication isogeny; it is a finite map for nonzero a , so one may speak of a degree-preserving pushforward $[a]_*$ in cohomology. Fix now an auxiliary integer $c > 1$: strictly speaking, $[a]$ does not restrict to a map on the torsion complement $T - T[c]$, but we nevertheless may do the standard trick (e.g., see [BCG1, §3]) of defining a pushforward for $(a, c) = 1$ on cohomology by composing with the restriction map

$$H^\bullet(T - T[c], \Omega_{T-T[c]}^n) \rightarrow H^\bullet(T - T[ac], \Omega_{T-T[ac]}^n) \xrightarrow{[a]_*} H^\bullet(T - T[c], \Omega_{T-T[c]}^n)$$

which we also denote, a bit abusively, by $[a]_*$. (Here, we write the cohomology groups most relevant to us, but this applies whenever one can pushforward-by-finite and pullback-by-flat.)

The following proposition summarizes the construction in equivariant coherent cohomology:

Proposition 2.3. *There is a unique cohomology class*

$$z_{\Gamma}^{(c)} \in H_{\Gamma}^1(T - T[c], \Omega_{T-T[c]}^2)^{(0)}$$

characterized (as indicated by the superscript (0)) by being invariant by $[a]_$ for all integers a relatively prime to c , and having residue*

$$[\delta_{T[c]-c^4\{0\}}] \in H_{\Gamma, T[c]}^2(T, \Omega_T^2)$$

along the residue map defined by (2.3).

Proof. In the long exact sequence in cohomology induced by (2.3), the relevant portion is

$$\dots \rightarrow H_{\Gamma}^1(T, \Omega_T^2) \rightarrow H_{\Gamma}^1(T - T[c], \Omega_{T-T[c]}^2) \rightarrow H_{T[c], \Gamma}^2(T, \Omega_T^2) \rightarrow H_{\Gamma}^2(T, \Omega_T^2) \rightarrow \dots$$

Note that to view currents on $T[c]$ inside the local cohomology group, we are using the pushforward map to the mapping cone defined before (2.3).

The image of $[\delta_{T[c]-c^4\{0\}}]$ is zero under the last map because we will construct, in Section 3, an explicit lift showing that $\delta_{T[c]-c^4\{0\}}$ is exact in the double complex $C^{\bullet}(T, \mathcal{D}_T^{2, \bullet})$. To obtain a unique trace-fixed lift, therefore, it suffices to show that $H_{\Gamma}^2(T, \Omega_T^2)^{(0)}$ vanishes. Indeed, the ‘‘Lieberman trick’’ argument (as employed in, e.g., [BCG1, §3.2]) applies verbatim: the Hochschild–Serre spectral sequence gives us

$$E_2^{p,q} := H^p(\Gamma, H^q(T, \Omega_T^2)) \Rightarrow H_{\Gamma}^{p+q}(T, \Omega_T^2)$$

functorial for the trace maps $[a]_*$. The argument of loc. cit., together with degeneration of the Hodge–de Rham spectral sequence for T (as a Kähler manifold), shows that this spectral sequence degenerates on the second page, and that $[a]_*$ acts by a^{2-q} on the term $H^p(\Gamma, H^q(T, \Omega_T^2))$. It immediately follows that $H_{\Gamma}^1(T, \Omega_T^2)^{(0)} = 0$. $\square$

Corollary 2.4. *If an element ω of the double complex $C^{\bullet}(\Gamma, \mathcal{D}_T^{2, \bullet})$ which is trace-fixed and has total differential $\delta_{T[c]-c^4\{0\}}$, then it represents the class*

$$z_{\Gamma}^{(c)} \in H_{\Gamma}^1(T - T[c], \Omega_{T-T[c]}^2).$$

Proof. This implies that the represented cohomology class will be trace-fixed and have image in the local cohomology group $H_{T[c], \Gamma}^2(T, \Omega_T^2)$ equal to the image of $[\delta_{T[c]-c^4\{0\}}]$ under the local pushforward

$$H^0(T[c], \Omega_{T[c]}^0) \rightarrow H_{T[c], \Gamma}^2(T, \Omega_T^2),$$

which are the two characterizing properties. $\square$

Constructing an ω as in the preceding corollary will be our first main goal in the next section 3.

Now, let $H \subset T$ be an arrangement elliptic hyperplanes in T containing $T[c]$, i.e. the image of some homomorphic inclusions of complex tori $E \rightarrow T$ over $\mathcal{H}$. We wish to restrict to the complement of the

Γ -orbit of H , “ $U := T - \Gamma \cdot H$ ”. However, this is not well-behaved since this orbit is always infinite: to make sense of this in cohomology, we define the equivariant cohomology groups (which we notate $H_\Gamma^\bullet(U, \Omega_U^2)$) by the double complex

$$C^\bullet(\Gamma, \varinjlim_S \mathcal{D}_{T-S}^{2,\bullet})$$

where the direct limit goes over all complements associated to finite subsets S of the components of $\Gamma \cdot H$; then one may restrict classes from $H_\Gamma^\bullet(T - T[c], \Omega_{T-T[c]}^2)$ “to U ” as if it were a well-behaved space. The advantage of doing this is:

Proposition 2.5. *The colimit $\varinjlim H^i(T - S, \Omega_{T-S}^2)$ is zero for any $i > 0$.*

Proof. Note that a punctured elliptic curve is an affine variety, and hence a Stein manifold; thus, removing two elliptic hyperplanes from a squared elliptic curve results in a Stein manifold as well. Then the $T - S$ are cofinally (i.e. “eventually” in the directed diagram) fiberwise Stein manifolds over the contractible Stein base $\mathcal{H}$, and thus the total space is also Stein. It results that coherent sheaves have vanishing higher cohomology. $\square$

As a result, the Hochschild-Serre spectral sequence for U (or rather the associated double complex which is our definition of “cohomology of U ”) gives rise to an edge map

$$e : H_\Gamma^1(U, \Omega_U^2) \rightarrow H^1(\Gamma, H^0(U, \Omega_U^2)) \cong H^1(\Gamma, \varinjlim_S H^0(T - S, \Omega_{T-S}^2))$$

where here we use the exactness of filtered colimits in the latter identification. We denote

$$Z_\Gamma^{(c)} \in H^1(\Gamma, H^0(U, \Omega_U^2))$$

as the resulting cohomology class valued in meromorphic forms.

This group cohomology class valued in 2-forms on (an open subspace of) the bundle T parameterizes a family of theta lifts: let ω be the dual Hodge bundle of T , a $\mathrm{GL}_2(R)$ -locally free sheaf on $\mathcal{H}$ which descends under quotients by arithmetic groups to the sheaf usually given the same name on modular curves. Then contracting with the $\mathrm{GL}_2(\mathbb{Z})$ -fixed vector field $\frac{\partial}{\partial z_1} \otimes \frac{\partial}{\partial z_2}$ gives a map

$$H^0(U, \Omega_U^2) \rightarrow H^0(U, \pi^* \omega).$$

Pullback by any Γ -fixed torsion section x disjoint from S , then yields our theta lift:

Definition 2.6. *We define the “theta lift at x ” cohomology class*

$$\Theta_{\Gamma,x}^{(c)} := x^*(Z_\Gamma^{(c)} \cap (\frac{\partial}{\partial z_1} \otimes \frac{\partial}{\partial z_2})) \in H^1(\Gamma, H^0(\mathcal{H}, \omega^{\otimes 2})),$$

which can be viewed via integration as a map from $H_1(\Gamma)$ to a weight-2 modular form inside $H^0(\mathcal{H}, \omega^{\otimes 2})$, as values of the integral will be fixed by any level structure H stabilizing the section x : all of the other above maps are $\mathrm{GL}_2(\mathbb{R})$ -equivariant, and pullback is H -invariant.

Note that S is not part of the notation above: for any given nonzero x , one can always find a disjoint $S \subset T$ invariant by the stabilizer of x , and this choice does not affect the end result: for example, simply pick the initial hyperplanes H to be a set of elliptic hyperplanes which pass through all c -torsion, but not x , which is possible by the Chinese remainder theorem. Given two different such arrangements, one can always restrict away from their union (while still preserving x in the complement), so it is clear the pulled back class is independent of the choice of removed sub-elliptic curves.

We conclude this section by summarizing the results of [BCG3] and the author's thesis [X] which show that the “coherent cohomology theta lift” we constructed above coincides with the analytic theta lift of [BCG1, §13] for $(\mathrm{GL}_2(\mathbb{Q}), \mathrm{GL}_2(\mathbb{Q}))$. The following theorem assembles the arguments of [BCG1, Theorem 29], [BCG3, Theorem 1.8, Theorem 2.8, Theorem 2.10], and [X, Propositions 3.8–3.13]:

Theorem 2.7. *There is a unique class $z_\Gamma^{dR,(c)} \in H_\Gamma^3(T - T[c], \mathbb{C})^{(0)}$ whose residue in $H_{\Gamma, T[c]}^4(T[c]) \cong H^0(T[c])^\Gamma$ is $[\delta_{T[c]-c^4\{0\}}]$ (where here we have used the excision isomorphism). The explicit c -stabilized trace-fixed differential 3-form $E_\psi^{(c)}$ of [BCG1, Theorem 21] defined by c -stabilizing their regularized theta series E_ψ via*

$$E_\psi^{(c)} := ([c]^* - c^4)E_\psi,$$

on the torus $T(\Gamma) := E\Gamma \times_\Gamma T$, represents this class,³ and its degree- $(1, 0, 2, 0)$ component⁴ $\theta_\psi^{(c)}$ is closed along $E\Gamma$, and the map

$$(2.5) \quad \gamma \mapsto \left(\int_{[\gamma]} x^* [\theta_\psi^{(c)} \cap (\frac{\partial}{\partial z_1} \otimes \frac{\partial}{\partial z_2})] \right)$$

agrees with $\Theta_{\Gamma, x}^{(c)}$.

Proof. As observed in loc. cit., the calculations of [BCG3, §9.3.3] imply that $E_\psi^{(c)}$ differs by a Γ -coboundary (a form exact for the differential on $E\Gamma$) and a form exact on T , having no contribution to the $(1, 0, 2, 0)$ component, from a form (which we may call $\theta_\psi'^{(c)}$) closed along Γ , whose integrals along geodesic 1-simplices $[\gamma]$ are valued in meromorphic forms on T with log poles along elliptic hyperplanes. It follows that

$$\left[\gamma \mapsto \left(\int_{[\gamma]} \theta_\psi^{(c)} \cap (\frac{\partial}{\partial z_1} \otimes \frac{\partial}{\partial z_2}) \right) \right] = \left[\gamma \mapsto \left(\int_{[\gamma]} \theta_\psi'^{(c)} \cap (\frac{\partial}{\partial z_1} \otimes \frac{\partial}{\partial z_2}) \right) \right] + (\text{a coboundary for } \Gamma)$$

represents the Hochschild–Serre edge map image of $z_\Gamma^{dR,(c)}$ in

$$(2.6) \quad H^1(\Gamma, H^2(U_{fin}))^{(0)} \cong H^1(\Gamma, H^2(U_{fin})^{(0)}) \cong H^1(\Gamma, H^0(\mathcal{H}, R^2\pi_*\mathbb{C}_{U_{fin}}^{(0)}))$$

where the first identification is by inductively applying the same Lieberman isogeny-projector trick used earlier to split the coefficients into trace-isotypic components (e.g., as used in [BCG3, §1.4.3]), and the second

³Here, as in [BCG3, Appendix A], one uses geodesic simplices to make an Γ -equivariant homology equivalence between the universal covering $E\Gamma$ with a symmetric space for $\mathrm{GL}_n(\mathbb{R})$; we implicitly use this equivariant-classifying space dictionary throughout without comment for brevity.

⁴Here, the indexing for degrees is along $E\Gamma$, $\mathcal{H}$, and holomorphic/antiholomorphic along T , respectively.

identification is from the Leray–Serre spectral sequence for the bundle $U_{fin} \rightarrow \mathcal{H}$ over the contractible base $\mathcal{H}$, which degenerates as shown thanks to the acyclicity of $\mathcal{H}$. Here, U_{fin} is as before the pro-space built out of all elliptic hyperplane complements.⁵

Further, since Bergeron–Charollois–Garcia show that the values of this class are meromorphic differentials with log poles, it lies in the image under the pushforward of sheaves

$$(2.7) \quad (\Omega_{log/\mathcal{H}}^2)^{(0)} \rightarrow R^2\pi_*\mathbb{C}_{U_{fin}}^{(0)},$$

where $\Omega_{log/\mathcal{H}}^2$ is the sheaf of *relative forms*⁶ over $\mathcal{H}$ with log poles along elliptic hyperplanes. This last map is actually an isomorphism, as shown stalkwise in [BCG3, Chapitre 3].

It then follows from a formal functorial argument that our edge map class $Z_\Gamma^{(c)}$ restricted to the group $H^1(\Gamma, H^0(U_{fin}, \Omega_{U_{fin}}^2))$ has the same image in $H^1(\Gamma, H^0(U_{fin}, \Omega_{U_{fin}/\mathcal{H}}^2))$ as the image of $z_\Gamma^{dR, (c)}$ under (2.6), composed with the inverse of (2.7), since both factor through the edge map image of the restriction to U_{fin} of the equivariant class in $H_\Gamma^1(T - T[c], \Omega_{(T-T[c])/\mathcal{H}}^2)$ characterized in the localization exact sequence

$$H_\Gamma^1(T, \Omega_{T/\mathcal{H}}^2) \rightarrow H_\Gamma^1(T - T[c], \Omega_{(T-T[c])/\mathcal{H}}^2) \xrightarrow{\text{res}} H_{\Gamma, T[c]}^2(T, \Omega_{T/\mathcal{H}}^2) \rightarrow H_\Gamma^2(T, \Omega_{T/\mathcal{H}}^2)$$

by the same two properties of trace-fixedness and having residue $[T[c] - c^4\{0\}] \in H^0(T[c], \mathcal{O}_{T[c]})^\Gamma \subset H_{\Gamma, T[c]}^2(T, \Omega_{T/\mathcal{H}}^2)$ (e.g., as written down in Kings–Sprang [KS, Definition 2.21], in a coefficient-twisted setting).

Since the original class $Z_\Gamma^{(c)}$ is valued in forms on a larger hyperplane complement $U \supset U_{fin}$ containing x , this implies that $\theta_\psi^{(c)}$ differs by an $E\Gamma$ -exact form from a form whose integrals along geodesic simplices are valued in meromorphic forms regular along x . The result follows by pulling back by x . $\square$

3. CONSTRUCTING THE LIFT

3.1. Kronecker-Eisenstein series as theta functions. We now know that in principle, one can compute Eisenstein classes by finding a suitable element in a distributional Dolbeault complex. Finding the currents which realize this element is essentially a cocycle lifting problem, which we tackle in this section.

Recall the Kronecker-Eisenstein series [Weil]

$$(3.1) \quad K_a(s, \tau, z, u) = \sum'_{\omega \in \mathbb{Z} + \mathbb{Z}\tau} \frac{(\overline{\omega + z})^a}{|\omega + z|^{2s}} \exp\left(2\pi i \frac{\omega \bar{u} - \bar{\omega} u}{\tau - \bar{\tau}}\right)$$

for $\tau \in \mathcal{H}$, $z, u \in \mathbb{C}$, and $a \in \mathbb{Z}$; the superscript apostrophe denotes that the sum omits any term where $\omega + z = 0$. This series is convergent for $\text{Re } s > 1 + a/2$, but has meromorphic continuation to all $s \in \mathbb{C}$, with simple poles possibly only at $s = 0$ (when $z \in \mathbb{Z} + \mathbb{Z}\tau$ and $a = 0$) $s = 1$ (when $u \in \mathbb{Z} + \mathbb{Z}\tau$ and $a = 0$). We

⁵As before, all associated cohomology groups should be understood as direct limits over restrictions over complements of finite hyperplane arrangements, which by exactness of filtered colimits are computed by directed limits over their associated de Rham complexes.

⁶The sheaves of relative forms $\Omega_{X/Y}^i$ for a submersion of complex manifolds $f : X \rightarrow Y$ is the quotient $\Omega_X^i / f^* \Omega_Y^i$.

define special notations, following [BCG1, §9], for the specializations we need:

$$(3.2) \quad E_1(\tau, z) := \frac{i}{2\pi} K_1(1, \tau, z, 0),$$

$$(3.3) \quad E_2(\tau, z) := -\frac{1}{4\pi y} K_2(1, \tau, z, 0).$$

Since these functions are manifestly $(\mathbb{Z} + \mathbb{Z}\tau)$ -periodic in z , we can consider $E_1(\tau, z)$ and $E_2(\tau, z)$ as functions on E , holomorphic in τ , which are smooth away from $z = 0$. Further, $E_1(\tau, z)$ is odd and $E_2(\tau, z)$ is even in z , for any τ ; in particular, we see that $E_1(\tau, 0) = 0$. As noted in [Weil], these two functions specialize at torsion points

$$E_1(\tau, \alpha - \beta\tau), E_2(\tau, \alpha - \beta\tau); (\alpha, \beta) \in (\mathbb{Q}/\mathbb{Z})^2 \setminus \{(0, 0)\}$$

to holomorphic modular forms in τ of weight 1, respectively 2, which are the classical Eisenstein series of those weights, of level corresponding to the stabilizer of (α, β) in $\mathrm{SL}_2(\mathbb{Z})$.

From [BCG1, (9.6)] together with [BCG1, Proposition 20], we observe that we have the distribution relations as *forms*

$$(3.4) \quad [a]_*(E_1(\tau, z) dz) = E_1(\tau, z) dz,$$

$$(3.5) \quad [a]_* E_2(\tau, z) = E_2(\tau, z)$$

for any integer $a \in \mathbb{Z} \setminus \{0\}$. Further, [BCG1, Theorem 19] and [BCG1, §9.4] together imply that, considered as a current on E ,

$$\bar{\partial} E_1(\tau, z) dz = \delta_0 - \mathrm{vol}_{E_\tau} = \delta_0 - \frac{2idz \wedge d\bar{z}}{y}$$

where here we give the formula for the normalized volume form on E . Away from the zero section (ignoring the current of integration δ_0), this derivative holds on the level of functions. Furthermore, [Weil, (35)] says that

$$\bar{\partial} E_2(\tau, z) = \frac{i}{2y} \cdot E_1(\tau, z) d\bar{z}$$

as functions (and hence currents) on E . Thus, both E_1 and E_2 are smooth away from the zero section, where E_2 is once-differentiable and E_1 has a singularity. Note that the formulas in [BCG1, §9.4] imply that on a punctured neighborhood of the zero section, E_1 looks like it has a simple log pole (even though its actual value at zero is 0). It therefore makes sense to consider forms like $E_1(\tau, z) dz$ and $E_2(\tau, z) dz$ as $(1, 0)$ -currents on E by considering them as kernels of integration as in the map ν_X we defined for smooth forms, since the corresponding integrals will converge for compactly supported smooth forms.

3.2. Explicit isogeny-fixed currents. We now construct a certain element of total degree 1 (or 3, if one adds in the Hodge weight 2)

$$\zeta \in C^\bullet(\mathrm{GL}_2(\mathbb{Q}), (\mathcal{D}_T^{2, \bullet})^{(0)})$$

whose total differential is the cocycle

$$\delta_0 - \mathrm{vol}_T \in Z^0(\mathrm{GL}_2(\mathbb{Q}), (\mathcal{D}_T^{2, 2})^{(0)}) = [(\mathcal{D}_T^{2, 2})^{(0)}]^{\mathrm{GL}_2(\mathbb{Q})}.$$

Here, the action of $\mathrm{GL}_2(\mathbb{Q})$ comes from the pushforward action of the monoid $M_2(\mathbb{Z}) \cap \mathrm{GL}_2(\mathbb{Z})$, extended to a group action using the trace-fixedness of scalar pushforward.

Lemma 3.1. *If we construct such a ζ , then for any integer $c > 1$, the restriction of $([c]^* - c^4)\zeta$ to any $\Gamma \subset \mathrm{GL}_2(\mathbb{Z})$ and $T - T[c] \subset T$ represents the class $z_\Gamma^{(c)}$.*

Proof. If the total differential of ζ is $\delta_0 - \mathrm{vol}_T$, then the total differential of $([c]^* - c^4)\zeta$ is $\delta_{T[c]} - c^4\delta_0$. Furthermore, it remains invariant by all isogenies prime to c ; hence by the discussion at the end of the previous section, the result follows. $\square$

To construct this lift, we must specify $\zeta^{0,1} \in C^0(\mathrm{GL}_2(\mathbb{Q}), (\mathcal{D}_T^{2,1})^{(0)})$ and $\zeta^{1,0} \in C^1(\mathrm{GL}_2(\mathbb{Q}), (\mathcal{D}_T^{2,0})^{(0)})$ such that $\bar{\partial}\zeta^{1,0} = \partial_{\mathrm{GL}_2(\mathbb{Q})}\zeta^{0,1}$ where the latter differential refers to the group coboundary map.

We therefore fix the choice

$$\zeta^{0,1} := E_1(\tau, z_1) dz_1 \delta_{z_2=0} + E_1(\tau, z_2) dz_2 \wedge \mathrm{vol}_{z_1=0}$$

whose image under $\bar{\partial}$ is can be computed as

$$(3.6) \quad \bar{\partial}\zeta^{0,1} = \bar{\partial}[E_1(\tau, z_1) dz_2 \delta_{z_2=0} + E_1(\tau, z_2) dz_2 \wedge \mathrm{vol}_{z_1=0}]$$

$$(3.7) \quad = (\delta_0 - \mathrm{vol}_{z_1=0} \delta_{z_2=0}) + (\mathrm{vol}_{z_1=0} \delta_{z_2=0} - \mathrm{vol}_{z_1=0} \wedge \mathrm{vol}_{z_2=0})$$

$$(3.8) \quad = \delta_0 - \mathrm{vol}_T$$

It remains therefore to find, for each $\gamma \in \mathrm{GL}_2(\mathbb{Q})$, a lift to $(\mathcal{D}_T^{2,0})^{(0)}$ of

$$(\gamma_* - 1)\zeta^{0,1} = (\gamma_* - 1)[E_1(\tau, z_2) dz_2 \delta_{z_2=0} + E_1(\tau, z_1) dz_1 \wedge \mathrm{vol}_{z_1=0}].$$

We observe that there are no more choices to be made; these lifts are unique:

Lemma 3.2. *The trace-fixed complex*

$$0 \rightarrow (\mathcal{D}_T^{2,0})^{(0)} \rightarrow (\mathcal{D}_T^{2,1})^{(0)} \rightarrow (\mathcal{D}_T^{2,2})^{(0)}$$

is left-exact.

Proof. This can be proven identically to [SV, Lemma 6.2.1], as the only trace-fixed cohomology of Ω_T^2 is in top degree, coming from the form yielding the fundamental class of the 4-torus T in the Hodge-de Rham spectral sequence.⁷ $\square$

⁷In fact, one can show using Fourier series that this complex is almost right-exact as well, except for one dimension of cohomology on the right. This is unnecessary for us, so we omit it.

To find formulas for these lifts for a fully general matrix γ is complicated if approached directly: the push-forward action on currents can yield expressions with arbitrarily many terms. Instead, we decompose our matrices to simplify the calculation.

3.3. Telescoping with the Bruhat decomposition. The main observation we need is that we have the “telescoping” relation

$$(3.9) \quad ((\gamma_2 \gamma_1)_* - 1)\zeta^{0,1} = (\gamma_2)_*((\gamma_1)_* - 1)\zeta^{0,1} + ((\gamma_2)_* - 1)\zeta^{0,1}$$

reducing the problem of finding a lift for the product of matrices to the problem for the individual matrices. Thus, the problem of finding lifts can be reduced to a set of generators of $\mathrm{GL}_2(\mathbb{Q})$.

We recall that the Bruhat decomposition says that

$$\mathrm{GL}_2(\mathbb{Q}) = BwB = \begin{pmatrix} * & * \\ 0 & * \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} * & * \\ 0 & * \end{pmatrix}$$

where B denotes the upper triangular Borel subgroup, and w the antidiagonal Weyl element. We can make this more explicit by writing

$$(3.10) \quad \gamma := \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} 1 & a \\ 0 & c \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & d/c \\ 0 & -\frac{\det \gamma}{c} \end{pmatrix}$$

$$(3.11) \quad = \begin{pmatrix} 1 & a/c \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & c \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & -\frac{d}{\det \gamma} \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -\frac{\det \gamma}{c} \end{pmatrix}$$

whenever $c \neq 0$, for an arbitrary matrix $\gamma \in \mathrm{GL}_2(\mathbb{Q})$. Here, we have further decomposed the appearing upper triangular matrices by factoring them into a diagonal times a unipotent matrix. When $c = 0$, our matrix is already in the Borel, so we can directly write

$$\begin{pmatrix} a & b \\ 0 & d \end{pmatrix} = \begin{pmatrix} 1 & b/d \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a & 0 \\ 0 & d \end{pmatrix}$$

The point of this is that w , diagonal, and unipotent matrices all act in very computable ways on $\zeta^{0,1}$, enabling us to find lifts:

- (1) All diagonal matrices act trivially on $\zeta^{0,1}$, by the isogeny properties of E_1 we proved earlier together with the fact that the volume form of a torus is isogeny-invariant. Hence the corresponding lifts are zero.

- (2) For the element w , we can write $(w_* - 1)\zeta^{0,1}$ as

$$\begin{aligned} E_1(\tau, z_2) dz_2 \delta_{z_1=0} + E_1(\tau, z_1) dz_1 \wedge \mathrm{vol}_{z_1=0} - E_1(\tau, z_1) dz_1 \delta_{z_2=0} - E_1(\tau, z_2) dz_2 \wedge \mathrm{vol}_{z_2=0} \\ = \bar{\partial}[E_1(\tau, z_1)E_1(\tau, z_2) dz_1 \wedge dz_2] \end{aligned}$$

which is a very simple lift.

(3) For a unipotent matrix γ_u with upper-right entry u , we find that the term with the current $\delta_{z_2=0}$ cancels in $((\gamma_u)_* - 1)\zeta^{0,1}$ (since the coordinate z_1 is fixed), and what remains is

$$(3.12) \quad \frac{2iu}{y} E_1(\tau, z_2) dz_1 \wedge dz_2 \wedge d\bar{z}_2 = \bar{\partial} [4uE_2(\tau, z_2) dz_1 \wedge dz_2]$$

Combining these with our explicit Bruhat decomposition, we find the following formulas: in the case when $c = 0$, we obtain

$$(3.13) \quad \zeta^{1,0}(\gamma) = \frac{4b}{d} E_2(\tau, z_2) dz_1 \wedge dz_2.$$

In the case when $c \neq 0$, we use the full Bruhat decomposition to obtain the following:

Proposition 3.3. *The unique lift $\zeta^{1,0}$ on a matrix $\gamma \in \mathrm{GL}_2(\mathbb{Q})$ is given by the expression⁸*

$$(3.14) \quad \left[\begin{pmatrix} a & 1 \\ c & 0 \end{pmatrix}_* \frac{4d}{c \det \gamma} E_2(\tau, z_2) + \frac{1}{c} \begin{pmatrix} 1 & a \\ 0 & c \end{pmatrix}_* E_1(\tau, z_1) E_1(\tau, z_2) + \frac{4a}{c} E_2(\tau, z_2) \right] dz_1 \wedge dz_2.$$

We therefore have obtained full formulas for ζ , whose c -stabilizations will yield the kernel classes $Z_\Gamma^{(c)}$ for the Eisenstein theta lift after restriction to $\mathrm{GL}_2(\mathbb{Z})$. For a general matrix γ , there does not appear to be a substantial further simplification of the preceding formulas, besides writing out the pushforwards as sums.

3.4. Specializations at torsion points. The actual “theta lift” for $(\mathrm{GL}_2, \mathrm{GL}_2)$ is generally taken to be valued after contraction pullback by torsion sections. Fix an arithmetic subgroup $\Gamma \subset \mathrm{GL}_2(\mathbb{Z})$; in order to compare our results with the cohomological theta lift, we will restrict ζ to Γ .

Remark 3.4. This arithmeticity hypothesis is only necessary to compare with the cohomological construction of the theta lift: one can still obtain pulled-back cocycles on, say, S -arithmetic subgroups for some set of inverted places S , so long as they fix some torsion sections (of order necessarily prime to the places in S). We do not write down this extension here, but the formulas can be obtained by our same methods.

In this context, we wish to consider the pullback of

$$Z_\Gamma^{(c)} \cap \left(\frac{\partial}{\partial z_1} \otimes \frac{\partial}{\partial z_2} \right) \in H^1(\Gamma, H^0(U, \pi^* \omega_U^{\otimes 2}))$$

by a Γ -fixed torsion cycle D disjoint from $T[c]$. Analogous to the case of a single torsion section, we will write

$$\Theta_{\Gamma, D}^{(c)} \in H^1(\Gamma, H^0(\mathcal{H}, \omega^{\otimes 2}))$$

for this image. We wish to interpret this composite in terms of the explicit double complex representative ζ ; the main issue is that currents cannot in general be pulled back by closed immersions. However, observe

⁸Notice that here, we have to introduce factors coming from the determinants of the matrices to go from pushforwards of forms to pushforwards of functions.

from the formulas that our elements ζ actually are smooth away from the Γ -orbit of $\{z_1 = 0\} \cup \{z_2 = 0\}$. We therefore set $H_\Gamma := \Gamma \cdot (\{z_1 = 0\} \cup \{z_2 = 0\})$, and conclude that the restriction of ζ to $T - H_\Gamma$ (again, in the inductive limit sense) lands in the image of the quasi-isomorphic inclusion

$$\varinjlim \mathcal{A}_{T-H}^{2,\bullet} \rightarrow \varinjlim \mathcal{D}_{T-H}^{2,\bullet}$$

where H ranges over finite subarrangements of H_Γ .

We now concern ourselves with the case that D is supported on N -torsion for an integer $N > 1$ relatively prime to c . From the preceding discussion, we can conclude:

Proposition 3.5. *We have the equality*

$$[D^*([c]^* - c^4)\zeta^{1,0} \cap (\frac{\partial}{\partial z_1} \otimes \frac{\partial}{\partial z_2})] = \Theta_{\Gamma,D}^{(c)} \in H^1(\Gamma, H^0(\mathcal{H}, \omega^{\otimes 2}))$$

so long as D is disjoint from H_Γ .

Proof. From the preceding discussion, ζ may be considered as an element of the double complex $C^\bullet(\Gamma, \mathcal{A}_{T-H_\Gamma}^{2,\bullet})$; thus, $([c] - c^4)\zeta^{1,0}$ is an element of $C^\bullet(\Gamma, \mathcal{A}_{T-[c]^{-1}H_\Gamma}^{2,\bullet})$ representing the class of $Z_\Gamma^{(c)}$ for $U = T - [c]^{-1}H_\Gamma$.

If D is disjoint from H_Γ , then it is also disjoint from $[c]^{-1}H_\Gamma$, since this is equivalent to $[c](D) \not\subset H_\Gamma$, and for any N -torsion section, any elliptic hyperplane must either contain the entire subgroup generated by it or be disjoint from it. Thus, the indicated pullback makes sense, and the result follows. $\square$

Note that if $c \equiv 1 \pmod{N}$, we can write the left-hand side as $(1 - c^4)[D^*\zeta^{1,0}]$.

The restriction on D is rather irritating, as it depends on the arbitrary choice of coordinates z_1, z_2 we used to choose our lift $\zeta^{0,1}$: for any given torsion section, we could simply start with a different lift to obtain formulas for the pullback. However, this would result in a somewhat unsatisfying lack of unity in our formulas. Luckily, we can use a trick to bypass this issue entirely, and make the formulas valid even for “bad” torsion sections:

Theorem 3.6. *Suppose $c \equiv 1 \pmod{N}$. Then for any N -torsion cycle D disjoint from the identity, we have that*

$$[D^*\zeta^{(1,0)}] = \frac{1}{1 - c^4} \Theta_{\Gamma,D}^{(c)} \in H^1(\Gamma, H^0(\mathcal{H}, \omega^{\otimes 2})).$$

where the pullback $D^*\zeta^{(1,0)}$ is always defined (even on the “bad hyperplanes” in H_Γ) using Lemma below, viewing $\zeta^{(1,0)}$ as a cocycle on Γ valued in M .

The proof consists of “bootstrapping” from torsion points disjoint from H_Γ to all of them. In order to do this, we will need the following lemma allowing us to “improve” our current-valued cocycles to be form-valued:

Lemma 3.7. *There is an injection*

$$M \hookrightarrow (\mathcal{A}_{T-H_\Gamma}^{2,0})^{(0)}$$

where M is defined to be the module of $(2, 0)$ -forms on $T - \{0\}$ spanned by the $\mathrm{GL}_2(\mathbb{Q})$ -orbit of $E_2(z_2) dz_1 \wedge dz_2$ and $E_1(\tau, z_1)E_1(\tau, z_2) dz_1 \wedge dz_2$, given by sending

$$\omega \mapsto \left(\eta \mapsto \int_T \omega \wedge \eta \right).$$

Proof of lemma. The only non-formal assertion here is that this map is injective. Notice that the depth of this lemma's assertion comes precisely from the discontinuities of the forms in the orbit of $E_1(\tau, z_1)E_1(\tau, z_2) dz_1 \wedge dz_2$ along codimension-1 sub-elliptic curves, which are excluded from $T - H_\Gamma$.⁹

We note the following property of $E_1(\tau, z_1)E_1(\tau, z_2) dz_1 \wedge dz_2$: suppose $x = (x_1, x_2) \in T$ is a section lying on one of the subcurves of discontinuity $S \subset T$ (so $x_1 = 0$ or $x_2 = 0$) but not equal to zero. Then take any small $v = (v_1, v_2) \in \mathbb{C}^2$ not parallel to the curve of discontinuity of x , so that $x \pm v$ does not lie in S . By the oddness of E_1 , we find that the average of the translates by $\pm v$ vanishes as we shrink v :

$$\lim_{\epsilon \rightarrow 0^+} \frac{1}{2} (E_1(\tau, z_1 + \epsilon v_1)E_1(\tau, z_2 + \epsilon v_2) + E_1(\tau, z_1 - \epsilon v_1)E_1(\tau, z_2 - \epsilon v_2)) = 0$$

as an equality of coefficients of $dz_1 \wedge dz_2$.

By moving this argument around by the general linear action, this applies to *any* nonzero point on a codimension 1 discontinuity stratum of a function in the orbit of $E_1(\tau, z_1)E_1(\tau, z_2) dz_1 \wedge dz_2$.

Now consider an arbitrary $\omega \in M$, and suppose that ω gives the trivial $(2, 0)$ -current when considered as a kernel of integration. Then, ω must be identically zero outside a finite union of sub-elliptic curves through the identity. Consider an arbitrary nonzero point x on one of these subcurves $S \subset T$, and pick some decomposition

$$\omega = \omega_1 + \omega_2$$

where ω_1 consists of a sum of terms in the orbit of $E_2(\tau, z_2) dz_1 \wedge dz_2$ or $E_1(\tau, z_1)E_1(\tau, z_2) dz_1 \wedge dz_2$ which *do not* have a discontinuity along S , and ω_2 consists of a sum of terms from the latter orbit which *do* have a discontinuity along S . Pick now some vector $v \in \mathbb{C}^2$ such that the line segment between the points $x \pm v$ intersects no other discontinuity locus of any term in ω_1 or ω_2 ; this is always possible since x is nonzero and there are only finitely many terms to consider (and hence subcurves to avoid).

Then we find that

$$(3.15) \quad \frac{1}{2} \left(\lim_{\epsilon \rightarrow 0^+} \omega_2|_{x+\epsilon v} + \omega_2|_{x-\epsilon v} \right) = \frac{1}{2} \left(\lim_{\epsilon \rightarrow 0^+} \omega|_{x+\epsilon v} - [\omega_1]|_{x+\epsilon v} + \omega|_{x-\epsilon v} - [\omega_1]|_{x-\epsilon v} \right)$$

$$(3.16) \quad = \frac{1}{2} \left(\lim_{\epsilon \rightarrow 0^+} -[\omega_1]|_{x+\epsilon v} - [\omega_1]|_{x-\epsilon v} \right)$$

⁹Indeed, to appreciate the delicacy, observe if we change $T - \{0\}$ to T in its statement, the statement becomes false: see [BG, Proposition 3.7] for an example of a relation between the weight-1 and weight-2 series everywhere except the zero section.

since ω is identically zero on a neighborhood of x in $T - S$. But ω_1 is continuous in a neighborhood of x by assumption, so this expression is just $-\omega_1|_{z=x}$. On the other hand, the average of the two limits we started with is zero from the preceding discussion, so we conclude that $\omega_1|_{z=x} = 0$. We also have $\omega_2|_{z=x} = 0$ because forms in the orbit of $E_1(\tau, z_1)E_1(\tau, z_2) dz_1 \wedge dz_2$ are zero along their discontinuity loci by construction. We hence conclude that $\omega|_{z=x} = 0$; since this applies to any nonzero point x , we conclude that ω is the zero form on $T - \{0\}$. This concludes the proof of injectivity. $\square$

Proof of theorem. Thanks to the lemma, we can consider $\zeta^{1,0}$ to be a cocycle valued in M . Let $T[N]'$ denote the primitive N -torsion, and let $S \subset T$ be a sub-elliptic curve which is not the vanishing locus of either z_1 or z_2 . Let $x_0 \in S[N]'$ be any point, and write $\Gamma_1(x_0) \subset \mathrm{GL}_2(\mathbb{Z})$ for its stabilizer. By construction, $x_0 \notin H_{\Gamma_1(x_0)}$, and so by the previous discussion,

$$(3.17) \quad [x_0^* \zeta_\tau] = \frac{1}{1 - c^4} \Theta_{\Gamma, x_0}^{(c)}$$

for any $c \equiv 1 \pmod{N}$. In fact, this formula holds for any section $x \in S[N]'$, since they are all stabilized by $\Gamma_1(x_0)$ as the Γ -action is by “elliptic blocks”.

We now observe that $z_{\mathrm{GL}_2(\mathbb{Z})}^{(c)} \cap \frac{\partial}{\partial z_1} \otimes \frac{\partial}{\partial z_2}$ induces, by restriction, a class

$$(3.18) \quad \Theta_{\mathrm{GL}_2(\mathbb{Z}), N}^{(c)} \in H_{\mathrm{GL}_2(\mathbb{Z})}^1(T[N]', H^0(\mathcal{H}, \omega^{\otimes 2})) \cong H^1(\mathrm{GL}_2(\mathbb{Z}), \mathrm{hom}(T[N]', H^0(\mathcal{H}, \omega^{\otimes 2})))$$

where the isomorphism comes from the fact that $T[N]'$ is a union of contractible spaces, causing the collapse of the Hochschild-Serre spectral sequence. By functoriality of this spectral sequence, for any cycle $D \subset T[N]'$ stabilized by $\Gamma(D)$, the image of $\Theta_{\mathrm{GL}_2(\mathbb{Z}), N}^{(c)}$ under the composite of restriction and evaluation

$$H^1(\mathrm{GL}_2(\mathbb{Z}), \mathrm{hom}(T[N]', \mathbb{C})) \xrightarrow{\mathrm{res}} H^1(\Gamma(D), \mathrm{hom}(T[N]', \mathbb{C})) \xrightarrow{D} H^1(\Gamma(D), \mathbb{C})$$

yields $\Theta_{\Gamma(D), D}^{(c)}$.

Observe that there is a $\mathrm{GL}_2(\mathbb{Z})$ -equivariant map

$$M \rightarrow \mathrm{hom}(T[N]', H^0(\mathcal{H}, \omega^{\otimes 2})), f dz_1 \wedge dz_2 \mapsto (x \mapsto f(x)).$$

We claim that the pushforward of $([c]^* - c^4)\zeta$ under this map can be identified with $\Theta_{\mathrm{GL}_2(\mathbb{Z}), N}^{(c)}$, which would then imply the desired result for arbitrary primitive N -torsion cycles.

Indeed, there is an isomorphism of $\mathrm{GL}_2(\mathbb{Z})$ -modules

$$\mathrm{hom}(T[N]', H^0(\mathcal{H}, \omega^{\otimes 2})) \rightarrow \mathrm{Ind}_{\Gamma_1(x_0)}^{\mathrm{GL}_2(\mathbb{Z})} \mathrm{hom}(S[N]', H^0(\mathcal{H}, \omega^{\otimes 2})), f \mapsto (\gamma \mapsto f \circ \gamma^{-1})$$

where $\mathrm{hom}(S[N]', \mathbb{C})$ has, naturally, a trivial action of $\mathrm{GL}_2(\mathbb{Z})$. Hence, by Shapiro’s lemma it suffices to show that $([c]^* - c^4)\zeta$ and $\Theta_{\mathrm{GL}_2(\mathbb{Z}), N}^{(c)}$ agree upon restriction to $\Gamma_1(x_0)$ under the quotient

$$\mathrm{hom}(T_\tau[N]', H^0(\mathcal{H}, \omega^{\otimes 2})) \twoheadrightarrow \mathrm{hom}(S[N]', H^0(\mathcal{H}, \omega^{\otimes 2}))$$

dual to the obvious inclusion. But this is precisely (3.17), which we have already established. Assembling these identifications together for all $N > 1$ yields the full theorem. $\square$

Thus, from (3.14), when specialized at any nonzero torsion sections (or combination thereof), yields the following formula for the Eisenstein theta lift of [BCG1]:

Theorem 3.8. *Let*

(3.19)

$$\theta[\gamma](\tau, z_1, z_2) := \begin{cases} \frac{4b}{d} E_2(\tau, z_2) & \text{if } c = 0, \text{ else} \\ \left(\begin{pmatrix} a & 1 \\ c & 0 \end{pmatrix}_* \frac{4d}{c \det \gamma} E_2(\tau, z_2) + \frac{1}{c} \begin{pmatrix} 1 & a \\ 0 & c \end{pmatrix}_* E_1(\tau, z_1) E_1(\tau, z_2) + \frac{4a}{c} E_2(\tau, z_2) \right) \end{cases}$$

Then given any Γ -fixed combination of nonzero torsion sections

$$D = \sum_i c_i [(u_i, v_i)],$$

if we write

$$(3.20) \quad \theta[\gamma](D) := \sum_i c_i \theta[\gamma](\tau, u_i, v_i),$$

we have

$$\Theta_{\Gamma, D}^{(c)}(\gamma) = \theta[\gamma](D) \in H^0(\mathcal{H}, \omega^{\otimes 2})^H$$

where H is any subgroup of $\mathrm{SL}_2(\mathbb{Z})$ fixing D under the action by Möbius transformations on $\mathcal{H}$. (In particular, as expected, our cocycles are valued in modular forms of weight 2, rather than arbitrary sections of $\omega^{\otimes 2}$.) Here, c_i are integer coefficients, and u_i and v_i are elements of $(\mathbb{Z}/N)^2 - \{(0, 0)\}$, thought of as N -torsion sections on E .

These formulas are workable, but the presence of the pushforward matrices (which can be evaluated as finite sums over preimage torsion sections) make them slightly unwieldy. Analogously to the classical setting of periods of Eisenstein series [X4], the first term

$$\sum_i c_i \theta[\gamma](\tau, u_i, v_i) = \frac{4d}{c \det \gamma} \sum_i c_i \left[\begin{pmatrix} a & 1 \\ c & 0 \end{pmatrix}_* E_2(\tau, z_2) \right]_{(u_i, v_i)}$$

can be simplified if we assume that $\gamma \in SL_2(\mathbb{Z})$:

$$(3.21) \quad \frac{4d}{c \det \gamma} \sum_i c_i \left[\begin{pmatrix} a & 1 \\ c & 0 \end{pmatrix}_* E_2(\tau, z_2) \right]_{(u_i, v_i)} = 4 \frac{d}{c} \sum_i c_i \sum_{j, k \in (\mathbb{Z}/c)^2} E_2 \left(\tau, u_i - \frac{a}{c}(v_i + (j, k)) \right)$$

$$(3.22) \quad = 4 \frac{d}{c} \sum_i c_i \sum_{j, k \in (\mathbb{Z}/c)^2} E_2 \left(\tau, \frac{1}{c}(v_i + (j, k)) \right)$$

$$(3.23) \quad = \frac{4d}{c} \sum_i c_i E_2(\tau, v_i)$$

$$(3.24)$$

Here, we use the distribution property of E_2 , along with the fact that γ^{-1} stabilizes D . This latter fact implies that for all $i \in I$,

$$av_i - cu_i = v_{\sigma(i)}$$

for some permutation σ of the index set I such that $c_i = c_{\sigma(i)}$ for all $i \in I$.

Unfortunately, we do not see a natural way to simplify the $E_1 E_1$ term in any generality, analogously to the classical formulas for Eisenstein periods we discussed in [X4].

Remark 3.9. Note that the value at $\tau = \infty$ theta lift θ yields precisely the classical formula for the weight-2 Eisenstein cocycle reproven in loc. cit; this follows immediately from the fact that at $\tau = \infty$, the series $E_i(\tau, z)$ degenerates to the periodic Bernoulli polynomial $\frac{B_i(z)}{2i}$: this is immediate from the description of both functions by Hecke regularized (analytic continuation in s), since it holds for s with large enough real part that the Fourier series are absolutely convergent.

Hence as expected, the (GL_2, GL_2) -Eisenstein theta lift's degeneration at a cusp yields the (GL_2, GL_1) theta lift (in the sense described in [BCG1, §13]).

Remark 3.10. Instead of a formula, one can obtain a more efficient "continued fraction" algorithm for computing the lifts for matrices in $SL_2(\mathbb{Z}) \subset GL_2(\mathbb{Q})$ by using its famous generators S and T and the recursion principle (3.9): this is presented for Eisenstein cocycles presented in [Scz1, §2.4], but works identically here by replacing the Bernoulli polynomials with our Eisenstein-Kronecker series. The outputs of this algorithm will coincide with the preceding formulas by uniqueness of $\zeta^{1,0}$, though this is not obvious.

4. SOME PROPERTIES AND EXTENSIONS

4.1. Hecke equivariance. As one expects for a theta lift, the cocycle θ satisfies a compatibility property between two kinds of Hecke operators: geometric Hecke operators coming from the variable τ in the upper half-plane with its H -action, and a cohomological Hecke action coming from the matrix action fiberwise. Using our algebraic approach, one could prove this compatibility analogously to the approach in [SV, §6].

However, since a form of Hecke compatibility was already proven in [BCG3, Théorème 2.8] for a closely related cocycle, it is much easier for us to simply to import this result using our already-proven comparison.

To fix ideas, in this section we will consider the restriction of $\theta_{\Gamma,x}^{(n)}$ to $\Gamma := \Gamma_1(N) \subset \mathrm{SL}_2(\mathbb{Z})$ for some integer $N > 1$, and a torsion section $x = (0, x_2) : \mathcal{H} \rightarrow E^2$ which descends to level $Y_1(N)$; i.e., we take also $H = \Gamma_1(N) \leq \mathrm{SL}_2(\mathbb{Z})$. The below approach can be applied to broader contexts, but in this article we will remain in this setting.

We recall the definition of two kinds of Hecke operators for GL_2 acting on $[\theta]$: a fiberwise action coming from group cohomology, and a geometric action coming from the Möbius action on τ . Write Δ for the monoid of rank-2 integral matrices which stabilize $(1, 0) \in (\mathbb{Z}/N)^2$ for the standard left representation, so that $\Gamma \subset \Delta$. Given any double Γ -coset in Δ , we can decompose it finitely as

$$\Gamma\alpha\Gamma = \bigcup_i \alpha_i\Gamma.$$

There are two different Δ -actions we will consider on the module $H^1(\Gamma, H^0(\mathcal{H}, \omega^{\otimes 2})^\Gamma)$, as well as on the free module on H -orbits of connected components of the torsion subgroup of T . The *fiberwise* action, whose associated Hecke operator we will denote by $T(\alpha)$ for the Γ -double coset generated by α , has γ act by

$$(|\det \gamma| \gamma^{-1})^*,$$

this choice made so that for $\gamma \in \mathrm{SL}_2(\mathbb{Z})$ it coincides with the pushforward we have heretofore been considering, and the “modular” action, whose Hecke operator we will denote by $\mathbf{T}(\alpha)$ for the H -double coset generated by α , has Δ -action which sends

$$\gamma \cdot (\tau, z_1, z_2) := \left(\frac{a\tau + b}{c\tau + d}, \frac{z_1}{c\tau + d}, \frac{z_2}{c\tau + d} \right).$$

Then for any double coset $\Gamma\alpha\Gamma$ (writing with the fiberwise notational convention), the action of $\Gamma\alpha\Gamma$ on 1-cocycles can be defined as in [RW] (or [BCG3, §2.2.1]) by sending a 1-cocycle $c : \Gamma \rightarrow M$ valued in a Δ -module M to

$$\gamma \mapsto \sum_i \alpha_i c(\gamma_i)$$

where γ_i is defined by the relation $\alpha_i \gamma = \gamma_i \alpha_{\sigma(i)}$ for some permutation σ of the representatives α_i . (Note that our conventions differ slightly from loc. cit, both here and for the pullback action of Δ ; these two changes result in the same Hecke action.)

On invariants (with the modular or fiberwise action), the action of $\Gamma\alpha\Gamma$ is simpler to define, sending an element $x \in M^\Gamma$ to

$$\sum_i \alpha_i x.$$

Then (4), (5) of [BCG3, Théorème 2.8] tell us that:

Proposition 4.1.

$$T(\alpha) \circ \theta[\gamma](D) = \theta[\gamma](T(\alpha)D), \mathbf{T}(\alpha) \circ \theta[\gamma](D) = \theta[\gamma](\mathbf{T}(\alpha)D)$$

for any $\alpha \in \Delta$, where here in the brackets the Hecke action is the one mentioned on the torsion sections, and $\theta[\gamma](D)$ is as in (3.20).

In particular, let T_p and $\mathbf{T}_p$ be the double coset operators associated to a prime p , consisting of all matrices in Δ with determinant p . When p is relatively prime to N , these form just a single double coset, else they may be a sum of multiple such operators.¹⁰

Write δ_p for the torsion cycle comprised of all p -torsion points $((\alpha_1, \beta_1), (\alpha_2, \beta_2)) \in T_\tau[p]$ such that (α_1, β_1) and (α_2, β_2) are linearly dependent over $\mathbb{Z}/p$. Then we can compute that for any auxiliary integer $c > 1$, we have

$$T_p(T[c] - c^4\{0\}) = \mathbf{T}_p(T[c] - c^4\{0\}) = ([c]^* - c^4)(\delta_p + p\{0\}).$$

Pulling back this equality by x (since this commutes with the pullback action of $\Delta!$), we obtain:

Proposition 4.2. *We have the equality $T_p\theta[\gamma](x) = \mathbf{T}_p\theta[\gamma](x)$ for all primes p , i.e. as a function of γ ,*

$$\theta[\gamma](x) : H_1(\Gamma_1(N), \mathbb{Z}) \rightarrow H^0(Y_1(N), \omega^{\otimes 2})$$

is equivariant for the Hecke subalgebra generated by $\{T_p\}_p$ for the fiberwise, respectively modular Hecke actions on source and target.

Note in particular that this includes the entire anemic Hecke algebra (all operators of level prime to N), but does not necessarily include U_p for p dividing N .

4.2. Relations between modular forms. We use the calculation of [BCG1, §13.3] to relate our cocycle's values to restrictions of Hilbert–Eisenstein series, and thereby get relations between L -values. We recall the setup in loc. cit., specialized to our case: let F be a real quadratic field, and write let U_p be the group of units of $\mathcal{O}_F$ with norm $+1$ which are $1 \pmod{p}$. Pick a generator u_p for this group, up to torsion. Let $\mathfrak{k}$ be an ideal, with coordinatization $\alpha : \mathfrak{k} \xrightarrow{\sim} \mathbb{Z}^2$ associated to an algebra embedding

$$\iota : \mathcal{O}_F \hookrightarrow M_2(\mathbb{Z}), u_p \mapsto \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma(p).$$

The map ι also induces a map

$$\iota_* : (F \otimes_{\mathbb{Q}} \mathbb{R})_+^\times / \mathbb{R}_+^\times \rightarrow \mathrm{GL}_2(\mathbb{R})^+ / \mathrm{SO}(2) \cdot \mathbb{R}_+^\times = \mathcal{H}$$

¹⁰Note that this is not a universal convention for p dividing the level; some authors use T_p for one of the primitive suboperators in this case (which we call U_p).

whose image g_t in the upper half-plane becomes a closed geodesic upon quotient by any arithmetic group; this is the geodesic is preserved by the action of the positive-norm units in $\mathcal{O}_F^\times$, viewed as a subgroup of $\mathrm{SL}_2(\mathbb{Z})$.

Let $1_{[\mathfrak{a}]}$ be the function on the narrow class group $\mathrm{Cl}^+(F)$ whose value is $\mathrm{sgn} \mathbf{N}t$ on the class $[(t)\mathfrak{k}]$ for $t \in F$, and otherwise zero; let $1_{[\mathfrak{a}]}^{-(p)}$ be its pullback to the ray class group of level p ; viewed as a function on the monoid of ideals, this will now vanish at ideals divisible by p . Then specializing [DDP, (51)], there is a diagonally restricted Hilbert-Eisenstein series $G_1(1, 1_{[\mathfrak{a}]}^{-(p)}) = G_1(1_{[\mathfrak{a}]}^{-(p)}, 1)$ of level $\Gamma_0(p)$, coming from the component of the Hilbert modular surface corresponding to the class of the inverse different $\mathfrak{d}^{-1}$, whose series expansion in $\tau \in \mathcal{H}$ is given by the continuation to $s = 0$ of

$$-\frac{d_F}{4\pi^2[\mathcal{O}_F^\times : U_p]} \sum_{\mathfrak{k} \in \mathrm{Cl}(F)} \mathbf{N}\mathfrak{k} \left(\sum_{\substack{(m,n) \in (\mathfrak{k} \times \mathfrak{k})/U_p \\ p \nmid m}} \frac{1}{(m\tau + n)|m\tau + n|^{2s}} \right).$$

Here, we use the notation as in loc. cit. with d_F the discriminant of F and an unwritten “norm” in the denominators, so that for example $m\tau + n$ means

$$(\sigma_1(m)\tau + \sigma_1(n))(\sigma_2(m)\tau + \sigma_2(n))$$

for the pair of real embeddings σ_1, σ_2 of F .

Meanwhile, the computation before [BCG1, Theorem 30] shows that

$$\theta[\iota(u_p)](x) = -2^{2s-2} \Gamma(s+1)^2 \pi^{-2} \mathrm{Im}(\tau)^{2s} \mathbf{N}\mathfrak{k} \sum_{(m,n) \in \mathfrak{k}^2/U_p} \frac{1}{(x + m\tau + n)|x + m\tau + n|^{2s}} \Big|_{s=0}$$

where $x \in (F/\mathfrak{k})^2$, and we use the same notation otherwise as above. Comparing these two formulas, we find that

$$G_1(1, 1_{[\mathfrak{a}]}^{-(p)}) = \frac{d_F}{p^2[\mathcal{O}_F^\times : U_p]} \sum_{\substack{x \in p^{-1}(\mathfrak{k} \times \mathfrak{k})/(\mathfrak{k} \times \mathfrak{k}) \\ x_1 \notin \mathfrak{k}}} \theta[\iota(u_p)](x).$$

We therefore conclude:

Theorem 4.3. *We have the equality of modular forms in $\Gamma_0(p)$*

$$G_1(1, 1_{[\mathfrak{a}]}^{-(p)}) = \frac{d_F}{p^2[\mathcal{O}_F^\times : U_p]} \sum_{x \in E[p]^2 - K_p^2} \left[\frac{4(a+d)}{c} E_2(\tau, x_2) + \frac{1}{c} \left\{ \begin{pmatrix} 1 & a \\ 0 & c \end{pmatrix}_* E_1(\tau, z_1) E_1(\tau, z_2) \right\}_{z=x} \right]$$

where $K_p \subset E[p]$ consists of all points fixed by $\Gamma_0(p)$.

The spectral coefficients of the right-hand side of this equality are periods of cusp forms f along modular symbols (via Rankin-Selberg formulas, cf. [Kato1, Theorem 4.6]) and of the left-hand side are partial L -values of f over F (see [DPV1, Theorem C]). This identity can thus be thought of packaging all these period identities together into one modular identity.

One can also make analogous computations for more general ψ of higher conductor, and obtain analogous formulas for general ray class characters in this manner.

4.3. CM elliptic curves. Suppose now that τ satisfies a quadratic equation with rational coefficients; then the corresponding elliptic curve E_τ has complex multiplication by an order $\mathcal{O}$ in $K := \mathbb{Q}(\tau)$. Then we can extend the action of $\mathrm{GL}_2(\mathbb{Z})$ on T_τ to an action of $\mathrm{GL}_2(\mathcal{O})$, and therefore the action of $\mathrm{GL}_2(\mathbb{Q})$ on the trace-fixed distributional de Rham complex to an action of $\mathrm{GL}_2(K)$. Note that in this case, we can take “trace-fixed” to include all isogenies built out of the “scalar” endomorphisms in $\mathcal{O}$, because (3.4) generalizes to these isogenies [BK, Proposition 1.1.6].

Since this latter group has a Bruhat decomposition

$$\mathrm{GL}_2(K) = B_K \begin{pmatrix} & 1 \\ 1 & \end{pmatrix} B_K$$

(where here B_K denotes the upper triangular Borel of this group) exactly as over $\mathbb{Q}$, the arguments of section 3 go through almost exactly as before:

Theorem 4.4. *The map given by*

$$(4.1) \quad \gamma \mapsto \theta_\tau[\gamma](z_1, z_2) := \begin{cases} \frac{4b}{d} E_2(\tau, z_2) & \text{if } c = 0, \text{ else} \\ \begin{pmatrix} a & 1 \\ c & 0 \end{pmatrix}_* \frac{4\bar{d}}{\bar{c} \cdot \det \gamma} E_2(\tau, z_2) + \frac{1}{c} \begin{pmatrix} 1 & a \\ 0 & c \end{pmatrix}_* E_1(\tau, z_1) E_1(\tau, z_2) + \frac{4\bar{a}}{\bar{c}} E_2(\tau, z_2) \end{cases}$$

is a cocycle for $\mathrm{GL}_2(K)$ valued in functions on $T_\tau - \{0\}$, whose specialization at torsion sections gives the theta lift of [BCG1, §13] for the dual pair $(\mathrm{GL}_2(K), \mathrm{GL}_1(K))$.

Proof. The proof is exactly identical to the proof of Theorem 3.8. □

This imaginary quadratic cocycle, along with higher-weight generalizations, was also constructed analytically in the work [BCG2]; essentially the same cocycle was also constructed by Ito [Ito] earlier. As described in [BCG1, §13.4], it follows from Ito’s computation that in this case we also have the L -value identity

$$(4.2) \quad x^* \theta_\tau[\gamma] = \pm(\alpha - \alpha') L(\gamma, 1; x).$$

where

$$\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathcal{O}_K)$$

is hyperbolic with $(a + d)^2 \notin \{0, 1, 4\}$, stabilizing x a torsion section of E_τ^2 , α and α' are the roots of $cX^2 + (d - a)X - b = 0$, and the L -function is given by

$$L(\gamma, s; x) = \sum'_{(m,n) \in (\mathcal{O}_K^2 + x)/\langle \gamma \rangle} \frac{\overline{Q(m, n)}}{|Q(m, n)|^{2s}},$$

for the quadratic form Q

$$Q(m, n) = (m\alpha + n)(m\alpha' + n).$$

REFERENCES

- [BCG1] Nicolas Bergeron, Pierre Charollois, and Luis Garcia. “Transgressions of the Euler class and Eisenstein cohomology of $\mathrm{GL}_N(\mathbb{Z})$.” In: *Japanese Journal of Mathematics* 15 (2020), pp. 311–379.
- [BCG2] Nicolas Bergeron, Pierre Charollois, and Luis E. García. “Eisenstein cohomology classes for GL_N over imaginary quadratic fields”. In: *Journal für die reine und angewandte Mathematik* 797 (2023), pp. 1–40.
- [BCG3] Nicolas Bergeron, Pierre Charollois, and Luis Garcia. *Cocycles de groupe pour GL_n et arrangements d’hyperplans*. Vol. 39. CRM Monograph Series. Providence, RI: American Mathematical Society, 2023.
- [BG] Lev A. Borisov and Paul E. Gunnells. “Toric modular forms and nonvanishing of L -functions.” In: *Journal für die reine und angewandte Mathematik* 539 (2001), pp. 149–165.
- [BK] Shinichi Kobayashi Kenichi Bannai. “Algebraic theta functions and p -adic interpolation of Eisenstein-Kronecker numbers.” In: *Duke Math Journal*. 153.2 (2010), pp. 229–295.
- [BPPS] Cecilia Busuioc et al. “Modular Symbols with Values in Beilinson-Kato Distributions”. In: *Transactions of the American Mathematical Society* (2025).
- [DDP] Henri Darmon, Samit Dasgupta, and Robert Pollack. “Hilbert modular forms and the Gross-Stark conjecture.” In: *Annals of Mathematics* 174 (2011), pp. 439–484.
- [Dem] Jean-Pierre Demailly. *Complex Analytic and Differential Geometry*. OpenContent book, available from the author’s website. 2012. URL: <https://www-fourier.ujf-grenoble.fr/~demailly/manuscripts/agbook.pdf>.
- [DPV1] Henri Darmon, Alice Pozzi, and Jan Vonk. “Diagonal restrictions of p -adic Eisenstein families.” In: *Mathematische Annalen* 379 (2021), pp. 503–548. URL: <http://www.math.mcgill.ca/darmon/pub/Articles/Research/73.DPV1/DPV1.pdf>.
- [Ito] Hiroshi Ito. “A function on the upper half-space which is analogous to the imaginary part of $\log \eta(z)$.” In: *Journal für die reine und angewandte Mathematik* 373 (1987), pp. 148–165.
- [Kato1] Kazuya Kato. “ p -adic Hodge theory and values of zeta functions of modular forms.” In: *Astérisque*. 295 (2004), pp. 117–290.
- [KS] Guido Kings and Johannes Sprang. “Eisenstein–Kronecker classes, integrality of critical values of Hecke L -functions and p -adic interpolation”. In: *Annals of Mathematics* 202.1 (2025), pp. 1–109.

REFERENCES

- [RW] Young Rhie and George Whaples. “Hecke operators in the cohomology of groups.” In: *Journal of the Mathematical Society of Japan*. 22 (1970), pp. 431–442.
- [Scz1] Robert Sczech. “Eisenstein cocycles for $GL_2(\mathbb{Q})$ and values of L -functions in real quadratic fields.” In: *Commentarii Mathematici Helvetici*. 67 (1992), pp. 363–382.
- [Skoda] Henri Skoda. “A Dolbeault lemma for temperate currents.” In: *Annales Henri Lebesgue*. 4 (2021), pp. 879–896.
- [SV] Romyar Sharifi and Akshay Venkatesh. “Eisenstein cocycles in motivic cohomology”. In: *Compositio Mathematica* 160.10 (2024), pp. 2407–2479.
- [Weil] André Weil. *Elliptic Functions According to Eisenstein and Kronecker*. Classics in Mathematics. Springer-Verlag, 1999.
- [X] Peter Xu. “Arithmetic Eisenstein theta lifts.” PhD thesis. McGill University, 2023.
- [X4] Peter Xu. *Classical periods of Eisenstein series and Bernoulli polynomials in the equivariant cohomology of a torus*. Preprint. 2024.

REFERENCES